Plastic Bottles

Product / Application	BIS (IS)	ISO	ASTM	DIN	Other Standards
PET Bottle (Packaged Drinking Water)	IS 14543; IS 15410:2025 (Plastic Bottles/Containers for Packaging of Natural Mineral Water and Packaged Drinking Water — Specification); IS 17803:2022 (Potable Water Bottles (Plastic, Reusable) — Specification); IS 12252:2017 (Polyalkylene Terephthalates (PET & PBT) for their Safe Use in Contact with Foodstuffs, Pharmaceuticals and Drinking Water); IS 10146:1982 (Polyethylene for its Safe Use in Contact with Foodstuffs, Pharmaceuticals and Drinking Water); IS 10910:1984 (Polypropylene for its Safe Use in Contact with Foodstuffs, Pharmaceuticals and Drinking Water); IS 10151:2019 (Polyvinyl Chloride (PVC) for its Safe Use in Contact with Foodstuffs, Pharmaceuticals and Drinking Water);	ISO 22000 (Food Safety Management); ISO 9001 (Manufacture of Food Contact Packaging); ISO 18601 (Packaging and Environment); ISO 12418-1 (PET bottle post-consumer recycling materials);	; ASTM D2659 (PET and PETG Thermoplastic Containers); ASTM D638 (Tensile Properties of Plastics); ASTM D2463 (Drop Impact Resistance of Blow-Molded Thermoplastic Containers); ASTM D2684 (Standard Test Method for Permeability of Thermoplastic Containers to Packaged Reagents or Proprietary Products)	DIN EN 15593 (Packaging — Management of Hygiene in Production of Packaging for Foodstuffs); DIN EN 1186 (Materials and Articles in Contact with Foodstuffs); DIN EN ISO 527 (Tensile Properties); DIN EN ISO 1628-5 (Viscosity Number of PET)	FDA 21 CFR 177 (Food Contact Plastics); EU 10/2011; NSF/ANSI 61 (Drinking Water System Components); NSF/ANSI 51 (Food Equipment Materials)
PET Bottle (Natural Mineral Water)	IS 13428; IS 15410:2025 (Plastic Bottles/Containers for Packaging of Natural Mineral Water and Packaged Drinking Water — Specification); IS 17803:2022 (Potable Water Bottles (Plastic, Reusable) — Specification); IS	ISO 22000 (Natural Mineral Water Packaging); ISO 18601 (Packaging and Environment); ISO 12418-1 (PET bottle post-consumer recycling materials);	; ASTM D2659 (PET and PETG Thermoplastic Containers); ASTM D2463 (Drop Impact Resistance of Blow-Molded Thermoplastic Containers); ASTM D638 (Tensile Properties of Plastics)	DIN EN 15593 (Packaging Hygiene Management for Foodstuffs); DIN EN 1186 (Food Contact Materials); DIN EN ISO 527 (Tensile Properties); DIN EN ISO 1628-5 (Viscosity Number of PET)	FDA 21 CFR 177; EU 10/2011; NSF/ANSI 61; European Bottled Water Association Guidelines

Product / Application	BIS (IS)	ISO	ASTM	DIN	Other Standards
	12252:2017 (Polyalkylene Terephthalates (PET & PBT) for their Safe Use in Contact with Foodstuffs, Pharmaceuticals and Drinking Water); IS 10146:1982 (Polyethylene for its Safe Use in Contact with Foodstuffs, Pharmaceuticals and Drinking Water); IS 10910:1984 (Polypropylene for its Safe Use in Contact with Foodstuffs, Pharmaceuticals and Drinking Water); IS 10151:2019 (Polyvinyl Chloride (PVC) for its Safe Use in Contact with Foodstuffs, Pharmaceuticals and Drinking Water);				
PET Bottle (Carbonated Soft Drinks); PET Bottle (Juices)	IS 12252 (Polyethylene Terephthalate (PET) for Safe Use in Contact with Foodstuffs, Pharmaceuticals and Drinking Water) (Food Contact); IS 14543 (Packaged Drinking Water Bottles, where applicable);;IS 15410; IS 17803; IS 10146; IS 10910; IS 10151	ISO 22000 (Beverage Packaging); ISO 18601 (Packaging and Environment); ISO 12418-1 (PET bottle post-consumer recycling materials);	ASTM D2659 (PET and PETG Thermoplastic Containers); ASTM D2463 (Drop Impact Resistance of Blow-Molded Thermoplastic Containers); ASTM D4169 (Performance Testing of Shipping Containers and Systems); ASTM D638 (Tensile Properties of Plastics)	DIN EN 1186 (Materials in Contact with Foodstuffs); DIN EN ISO 527 (Tensile Properties); DIN EN ISO 1628-5 (PET Viscosity Number); DIN EN ISO 179-1 (Impact Strength)	FDA 21 CFR 177; EU 10/2011; NSF/ANSI 51; Coca-Cola Supplier Packaging Standards (Industry Requirement, where applicable)
PET Bottle (Edible Oil)	IS 12252 (Polyethylene Terephthalate (PET) for Safe Use in Contact with Foodstuffs, Pharmaceuticals and Drinking Water) (Food Contact);IS 15410; IS 17803; IS 10146; IS 10910; IS	ISO 12418-1 (PET bottle post-consumer recycling materials);	ASTM D2659 (PET and PETG Thermoplastic Containers); ; ASTM D543 (Resistance of Plastics to Chemical Reagents); ASTM D638 (Tensile Properties of Plastics)	DIN EN 1186 (Food Contact Materials); DIN EN ISO 527 (Tensile Properties); DIN EN ISO 175 (Resistance to Chemical Liquids); DIN EN ISO 1628-5 (PET Viscosity Number)	FDA 21 CFR 177; EU 10/2011; NSF/ANSI 51; JIS Z 1707 (Packaging Materials, where applicable)

Product / Application	BIS (IS)	ISO	ASTM	DIN	Other Standards
	10151				
PET Bottle (Milk)	IS 12252 (Polyethylene Terephthalate (PET) for Safe Use in Contact with Foodstuffs, Pharmaceuticals and Drinking Water) (Food Contact);IS 15410; IS 17803; IS 10146; IS 10910; IS 10151	ISO 12418-1 (PET bottle post-consumer recycling materials);	ASTM D2659 (PET and PETG Thermoplastic Containers); ASTM D4169 (Performance Testing of Shipping Containers and Systems); ASTM D543 (Resistance of Plastics to Chemical Reagents); ASTM D638 (Tensile Properties of Plastics)	DIN EN 1186 (Materials in Contact with Foodstuffs); DIN EN ISO 527 (Tensile Properties); DIN EN ISO 175 (Chemical Resistance); DIN EN ISO 1628-5 (PET Viscosity Number)	FDA 21 CFR 177; EU 10/2011; NSF/ANSI 51; 3-A Sanitary Standards (Dairy Food Contact Applications, where applicable)
PET Bottle (Pharmaceutical Syrup)	IS 12252 (Polyethylene Terephthalate (PET) for Safe Use in Contact with Foodstuffs, Pharmaceuticals and Drinking Water) (Pharmaceutical Contact);IS 15410; IS 17803; IS 10146; IS 10910; IS 10151	ISO 15378 (Primary Packaging Materials for Medicinal Products); ISO 9001 (Pharmaceutical Packaging Manufacture); ISO 12418-1 (PET bottle post-consumer recycling materials);	ASTM D2659 (PET and PETG Thermoplastic Containers); ASTM D4169 (Performance Testing of Shipping Containers and Systems); ASTM D638 (Tensile Properties of Plastics); ASTM D543 (Resistance of Plastics to Chemical Reagents)	DIN EN ISO 15378 (Primary Packaging Materials for Medicinal Products); DIN EN ISO 527 (Tensile Properties); DIN EN ISO 175 (Chemical Resistance); DIN EN ISO 1628-5 (PET Viscosity Number)	USP <661> (Plastic Packaging Systems); USP Class VI; FDA 21 CFR 177; EU Pharmacopoeia Requirements
PET Bottle (Oral Liquid)	IS 12252 (Polyethylene Terephthalate (PET) for Safe Use in Contact with Foodstuffs, Pharmaceuticals and Drinking Water) (Pharmaceutical Contact);IS 15410; IS 17803; IS 10146; IS 10910; IS 10151	ISO 15378 (Primary Packaging Materials for Medicinal Products); ISO 9001 (Pharmaceutical Packaging Manufacture); ISO 12418-1 (PET bottle post-consumer recycling materials);	ASTM D2659 (PET and PETG Thermoplastic Containers); ASTM D2463 (Drop Impact Resistance of Blow-Molded Thermoplastic Containers); ASTM D638 (Tensile Properties of Plastics); ASTM D256 (Izod Impact Resistance of Plastics)	DIN EN ISO 15378 (Primary Packaging Materials for Medicinal Products); DIN EN ISO 527 (Tensile Properties); DIN EN ISO 175 (Chemical Resistance); DIN EN ISO 1628-5 (PET Viscosity Number)	USP <661>; USP Class VI; FDA 21 CFR 177; European Pharmacopoeia Packaging Requirements
PET Bottle (Cosmetics)	IS 12252 (Polyethylene Terephthalate (PET) for Safe Use in Contact with Foodstuffs, Pharmaceuticals and Drinking Water) (Cosmetic Packaging Material);IS 15410; IS 17803; IS 10146; IS 10910; IS 10151	ISO 22716 (Cosmetics — Good Manufacturing Practices); ISO 18601 (Packaging and the Environment); ISO 12418-1 (PET bottle post-consumer recycling materials);	ASTM D2659 (PET and PETG Thermoplastic Containers); ASTM D638 (Tensile Properties of Plastics); ASTM D256 (Izod Impact Resistance of Plastics); ASTM D4169 (Performance Testing of Shipping Containers and Systems)	DIN EN ISO 527 (Tensile Properties); DIN EN ISO 175 (Chemical Resistance); DIN EN ISO 1628-5 (PET Viscosity Number); DIN EN ISO 179-1 (Impact Strength)	ISO 22716 (Cosmetic GMP — excluded ISO, therefore not listed); FDA 21 CFR 700 Series (Cosmetic Packaging Regulations); EU Cosmetic Regulation (EC) No. 1223/2009
PET Bottle (Personal Care Products)	IS 12252 (Polyethylene Terephthalate (PET) for Safe Use in Contact with Foodstuffs,	ISO 22716 (Cosmetics / Personal Care Packaging); ISO 18601 (Packaging and the Environment); ISO 12418-1 (PET bottle post-consumer recycling materials);	ASTM D2659 (Column Crush Properties of Blown Thermoplastic Containers); ASTM D2463 (Drop Impact Resistance of Blow-Molded Thermoplastic Containers); ASTM D2911/D2911M (Dimensions and	DIN EN ISO 527 (Tensile Properties of Plastics); DIN EN ISO 175 (Effect of Liquid Chemicals on Plastics); DIN EN ISO 1628-5 (Viscosity Number of Polyethylene Terephthalate); DIN EN ISO 179-1	FDA Cosmetic Regulations; EU Cosmetic Regulation (EC) No. 1223/2009; IKW Guidelines (German Cosmetic Industry Guidelines)

Product / Application	BIS (IS)	ISO	ASTM	DIN	Other Standards
	Pharmaceuticals and Drinking Water) (PET Material);IS 4985; IS 13592; IS 15778; IS 12231; IS 12818; IS 14182; IS 16087; IS 4984; IS 14333; IS 14885; IS 16088; IS 15225; IS 15265; IS 15801; IS 16098; IS 14402; IS 7834; IS 14735; IS 17546; IS 10124; IS 13593; IS 8008; IS 17424; IS 12235; IS 7634; IS 5382		Tolerances for Plastic Bottles); ASTM D638 (Tensile Properties of Plastics – supporting material test); ASTM D543 (Resistance of Plastics to Chemical Reagents – where chemical resistance is relevant)	(Charpy Impact Strength of Plastics)	
PET Bottle (Agrochemicals); PET Bottle (Industrial Chemicals)	IS 12252 (Polyalkylene Terephthalates (PET & PBT) for their Safe Use in Contact with Foodstuffs, Pharmaceuticals and Drinking Water); IS 14543 (Packaging Applications, where applicable);IS 15410; IS 17803; IS 10146; IS 10910; IS 10151	ISO 2230 (Packaging for Rubber & Plastics Products – Storage Guidance); ISO 16103 (Packaging for Dangerous Goods, where applicable); ISO 12418-1 (PET bottle post-consumer recycling materials);	ASTM D2659 (PET and PETG Thermoplastic Containers); ASTM D543 (Resistance of Plastics to Chemical Reagents); ASTM D1693 (Environmental Stress-Cracking of Ethylene Plastics, where PE components apply); ASTM D638 (Tensile Properties of Plastics);	DIN EN ISO 527 (Tensile Properties of Plastics); DIN EN ISO 175 (Resistance to Chemical Agents); DIN EN ISO 1628-5 (Viscosity Number of PET); DIN EN ISO 1183-1 (Density of Plastics)	UN Recommendations on the Transport of Dangerous Goods (UN Packaging Codes); EPA Packaging Requirements (where applicable); ADR Packaging Requirements (Dangerous Goods Transport)
PET Bottle (Laboratory Reagents)	IS 12252 (Polyalkylene Terephthalates (PET & PBT) for their Safe Use in Contact with Foodstuffs, Pharmaceuticals and Drinking Water);IS 15410; IS 17803; IS 10146; IS 10910; IS 10151	ISO 4796-1 (Laboratory Glass Bottles – application reference for laboratory reagent containers); ISO 15378 (Primary Packaging, where pharmaceutical reagents are involved); ISO 12418-1 (PET bottle post-consumer recycling materials);	ASTM D2659 (PET and PETG Thermoplastic Containers); ASTM D543 (Resistance of Plastics to Chemical Reagents); ASTM D638 (Tensile Properties of Plastics); ASTM D256 (Izod Impact Resistance of Plastics); ASTM D2684 (Standard Test Method for Permeability of Thermoplastic Containers to Packaged Reagents or Proprietary Products)	DIN EN ISO 527 (Tensile Properties); DIN EN ISO 175 (Chemical Resistance of Plastics); DIN EN ISO 1628-5 (PET Viscosity Number); DIN EN ISO 1183-1 (Density)	USP <661> (Plastic Packaging Systems); ISO 17025 laboratory requirements excluded; UL 94 (Plastic Fire Classification, where applicable); Chemical Resistance Guidelines from manufacturer specifications
HDPE Bottle (Milk)	IS 7328 (High Density Polyethylene Materials for Safe Use in Contact with Foodstuffs, Pharmaceuticals and Drinking Water) (Food Contact); IS 10146:1982 (Polyethylene for its Safe Use in Contact with Foodstuffs, Pharmaceuticals and Drinking Water);IS 15410; IS 17803; IS	ISO 22000 (Milk Packaging); ISO 18601 (Packaging and the Environment); ISO 12418-1 (PET bottle post-consumer recycling materials); ISO 13106 (Blow-moulded PP containers and bottle necks)	ASTM D4976 (Polyethylene Plastics Molding and Extrusion Materials); ASTM D256 (Izod Impact Resistance of Plastics); ASTM D638 (Tensile Properties of Plastics); ASTM D1693 (Environmental Stress-Cracking of Ethylene Plastics); ASTM D2561 (Standard Test Method for Environmental Stress-Crack Resistance of Blow-Molded Polyethylene Containers); ASTM D4504 (Standard Specification for Molded Polyethylene Open-Head Pails for Industrial Shipping of Nonhazardous Goods)	DIN EN 1186 (Materials and Articles in Contact with Foodstuffs); DIN EN ISO 527 (Tensile Properties); DIN EN ISO 1628-3 (Viscosity Number of Polyethylene); DIN EN ISO 1183-1 (Density of Plastics)	FDA 21 CFR 177 (Food Contact Plastics); EU 10/2011 (Food Contact Materials); 3-A Sanitary Standards (Dairy Applications); NSF/ANSI 51

Product / Application	BIS (IS)	ISO	ASTM	DIN	Other Standards
	12252; IS 10910; IS 10151				
HDPE Bottle (Detergent); HDPE Bottle (Lubricating Oil)	IS 7328 (HDPE Material); IS 14636 (Packaging Applications); IS 15410; IS 17803; IS 12252; IS 10146; IS 10910; IS 10151	ISO 18601 (Packaging and the Environment); ISO 9001 (Manufacture of Packaging Products); ISO 12418-1 (PET bottle post-consumer recycling materials); ISO 13106 (Blow-moulded PP containers and bottle necks)	ASTM D4976 (Polyethylene Plastics Molding and Extrusion Materials); ASTM D543 (Resistance of Plastics to Chemical Reagents); ASTM D1693 (Environmental Stress-Cracking of Ethylene Plastics); ASTM D638 (Tensile Properties of Plastics); ASTM D2561 (Standard Test Method for Environmental Stress-Crack Resistance of Blow-Molded Polyethylene Containers); ASTM D4504 (Standard Specification for Molded Polyethylene Open-Head Pails for Industrial Shipping of Nonhazardous Goods)	DIN EN ISO 527 (Tensile Properties); DIN EN ISO 175 (Chemical Resistance); DIN EN ISO 1628-3 (Viscosity Number of Polyethylene); DIN EN ISO 1183-1 (Density)	UN Dangerous Goods Packaging Requirements (where hazardous chemicals apply); DOT 49 CFR Packaging Requirements; ADR Packaging Regulations
HDPE Bottle (Bleach); Fluorinated HDPE Bottle	IS 7328 (HDPE Material); IS 14636 (Chemical Packaging Applications); IS 15410; IS 17803; IS 12252; IS 10146; IS 10910; IS 10151	ISO 16103 (Packaging for Dangerous Goods); ISO 18601 (Packaging and the Environment); ISO 12418-1 (PET bottle post-consumer recycling materials); ISO 13106 (Blow-moulded PP containers and bottle necks)	ASTM D543 (Resistance of Plastics to Chemical Reagents); ASTM D4976 (Polyethylene Plastics Molding and Extrusion Materials); ASTM D1693 (Environmental Stress-Cracking of Ethylene Plastics); ASTM D638 (Tensile Properties of Plastics); ASTM D2561 (Standard Test Method for Environmental Stress-Crack Resistance of Blow-Molded Polyethylene Containers); ASTM D4504 (Standard Specification for Molded Polyethylene Open-Head Pails for Industrial Shipping of Nonhazardous Goods)	DIN EN ISO 527 (Tensile Properties); DIN EN ISO 175 (Resistance to Chemical Liquids); DIN EN ISO 1628-3 (Viscosity Number of Polyethylene); DIN EN ISO 4892-3 (Artificial Weathering, where applicable)	UN Packaging Codes for Dangerous Goods; EPA Chemical Container Requirements; DOT 49 CFR Transport Packaging
HDPE Bottle (Pesticides)	IS 7328 (HDPE Material); IS 8190 (Pesticide Packaging Applications); IS 15410; IS 17803; IS 12252; IS 10146; IS 10910; IS 10151	HDPE Bottle (Pesticides);ISO 19932 (Packaging of Pesticides); ISO 18601 (Packaging and the Environment); ISO 12418-1 (PET bottle post-consumer recycling materials); ISO 13106 (Blow-moulded PP containers and bottle necks)	ASTM D543 (Resistance of Plastics to Chemical Reagents); ASTM D4976 (Polyethylene Plastics Molding and Extrusion Materials); ASTM D1693 (Environmental Stress-Cracking of Ethylene Plastics); ASTM D638 (Tensile Properties of Plastics); ASTM D2561 (Standard Test Method for Environmental Stress-Crack Resistance of Blow-Molded Polyethylene Containers); ASTM D4504 (Standard Specification for Molded Polyethylene Open-Head Pails for Industrial Shipping of Nonhazardous Goods)	DIN EN ISO 527 (Tensile Properties); DIN EN ISO 175 (Chemical Resistance); DIN EN ISO 1628-3 (Viscosity Number of Polyethylene); DIN EN ISO 1183-1 (Density)	UN Recommendations on Transport of Dangerous Goods; EPA Pesticide Container Requirements; CropLife International Packaging Guidelines
HDPE Bottle (Pharmaceutical)	IS 7328 (HDPE Material for Pharmaceutical Contact); IS 10146:1982 (Polyethylene for its Safe Use in Contact with Foodstuffs,	ISO 9001 (Pharmaceutical Packaging Manufacture); ISO 12418-1 (PET bottle post-consumer recycling materials); ISO 13106 (Blow-moulded PP containers and bottle necks)	ASTM D4976 (Polyethylene Plastics Molding and Extrusion Materials); ASTM D543 (Resistance of Plastics to Chemical Reagents); ASTM D638 (Tensile Properties of Plastics); ASTM D2463 (Drop Impact Resistance of Blow-Molded Thermoplastic Containers); ASTM D2561 (Standard	DIN EN ISO 15378 (Primary Packaging Materials for Medicinal Products); DIN EN ISO 527 (Tensile Properties); DIN EN ISO 175 (Chemical Resistance); DIN EN ISO 1183 (Density of Plastics)	USP <661> (Plastic Packaging Systems and Materials); USP Class VI (Biocompatibility, where applicable); FDA 21 CFR 177

Product / Application	BIS (IS)	ISO	ASTM	DIN	Other Standards
	Pharmaceuticals and Drinking Water);IS 15410; IS 17803; IS 12252; IS 10910; IS 10151		Test Method for Environmental Stress-Crack Resistance of Blow-Molded Polyethylene Containers); ASTM D4504 (Standard Specification for Molded Polyethylene Open-Head Pails for Industrial Shipping of Nonhazardous Goods)		
HDPE Bottle (Food Products)	IS 7328 (HDPE Material for Food Contact); IS 10146:1982 (Polyethylene for its Safe Use in Contact with Foodstuffs, Pharmaceuticals and Drinking Water);IS 4985; IS 13592; IS 15778; IS 12231; IS 12818; IS 14182; IS 16087; IS 4984; IS 14333; IS 14885; IS 16088; IS 15225; IS 15265; IS 15801; IS 16098; IS 14402; IS 7834; IS 14735; IS 17546; IS 10124; IS 13593; IS 8008; IS 17424; IS 12235; IS 7634; IS 5382	ISO 22000 (Food Packaging); ISO 18601 (Packaging and the Environment); ISO 12418-1 (PET bottle post-consumer recycling materials); ISO 13106 (Blow-moulded PP containers and bottle necks)	ASTM D2561 (Environmental Stress-Crack Resistance of Blow-Molded Polyethylene Containers); ASTM D2659 (Column Crush Properties of Blown Thermoplastic Containers); ASTM D2463 (Drop Impact Resistance of Blow-Molded Thermoplastic Containers); ASTM D2911/D2911M (Dimensions and Tolerances for Plastic Bottles); ASTM D4976 (Polyethylene Plastics Molding and Extrusion Materials – material classification); ASTM D792 (Density and Specific Gravity of Plastics); ASTM D638 (Tensile Properties of Plastics – supporting material test)	DIN EN 1186 (Materials and Articles in Contact with Foodstuffs); DIN EN ISO 527 (Tensile Properties); DIN EN ISO 1628-3 (Viscosity Number of Polyethylene); DIN EN ISO 1183-1 (Density)	FDA 21 CFR 177; EU 10/2011; NSF/ANSI 51; LFGB Food Contact Requirements
PP Bottle (Laboratory)	IS 10910 (Polypropylene and its Copolymers for Safe Use in Contact with Foodstuffs, Pharmaceuticals and Drinking Water);IS 15410; IS 17803; IS 12252; IS 10146; IS 10151	ISO 4796-1 (Laboratory Bottles); ISO 13485 (Laboratory / Medical Applications, where applicable); ISO 13106 (Blow-moulded PP containers and bottle necks)	ASTM D4101 (Polypropylene Injection and Extrusion Materials); ASTM D543 (Resistance of Plastics to Chemical Reagents); ASTM D638 (Tensile Properties of Plastics); ASTM D256 (Izod Impact Resistance of Plastics)	DIN EN ISO 527 (Tensile Properties); DIN EN ISO 1873-1 (Polypropylene Moulding and Extrusion Materials); DIN EN ISO 175 (Chemical Resistance); DIN EN ISO 1183-1 (Density)	USP <661> (Plastic Packaging Systems); USP Class VI (where medical use applies); FDA 21 CFR 177
PP Bottle (Pharmaceutical)	IS 10910 (Polypropylene and its Copolymers for Safe Use in Contact with Foodstuffs, Pharmaceuticals and Drinking Water) (Pharmaceutical Contact);IS 15410; IS 17803; IS 12252; IS 10146; IS 10151	ISO 15378 (Primary Packaging Materials for Medicinal Products); ISO 9001 (Pharmaceutical Packaging Manufacture); ISO 13106 (Blow-moulded PP containers and bottle necks)	ASTM D4101 (Polypropylene Injection and Extrusion Materials); ASTM D638 (Tensile Properties of Plastics); ASTM D543 (Resistance of Plastics to Chemical Reagents); ASTM D2463 (Drop Impact Resistance of Blow-Molded Thermoplastic Containers)	DIN EN ISO 15378 (Primary Packaging Materials for Medicinal Products); DIN EN ISO 527 (Tensile Properties); DIN EN ISO 175 (Chemical Resistance); DIN EN ISO 1873-1 (Polypropylene Materials)	USP <661>; USP Class VI; FDA 21 CFR 177; FDA Drug Packaging Requirements
PP Bottle (Chemical Reagents)	IS 10910 (Polypropylene for its Safe Use in Contact with Foodstuffs,	ISO 16103 (Packaging for Dangerous Goods, where applicable); ISO 9001 (Chemical Packaging Manufacture); ISO 13106 (Blow-moulded PP	ASTM D4101 (Polypropylene Injection and Extrusion Materials); ASTM D543 (Resistance of Plastics to Chemical Reagents); ASTM D638 (Tensile Properties of Plastics); ASTM D1693	DIN EN ISO 527 (Tensile Properties); DIN EN ISO 175 (Chemical Resistance); DIN EN ISO 1873-1 (Polypropylene Materials); DIN EN ISO 179-1 (Impact	UN Dangerous Goods Packaging Requirements; DOT 49 CFR Packaging Regulations; EPA Chemical Packaging Guidelines

Product / Application	BIS (IS)	ISO	ASTM	DIN	Other Standards
	Pharmaceuticals and Drinking Water);IS 15410; IS 17803; IS 12252; IS 10146; IS 10151	containers and bottle necks)	(Environmental Stress-Cracking of Ethylene Plastics, where applicable); ASTM D2684 (Standard Test Method for Permeability of Thermoplastic Containers to Packaged Reagents or Proprietary Products)	Strength)	
Multilayer Barrier Bottle	IS 14636 (Flexible Packaging for Food Products) (Barrier Packaging Applications); IS 15609 (Multilayer Plastic Pouches for Packaged Drinking Water) (Multilayer Packaging Applications); IS 15410; IS 17803; IS 12252; IS 10146; IS 10910; IS 10151	ISO 16103 (Packaging for Dangerous Goods); ISO 18601 (Packaging and the Environment); ISO 12418-1 (PET bottle post-consumer recycling materials); ISO 13106 (Blow-moulded PP containers and bottle necks)	ASTM D3985 (Oxygen Gas Transmission Rate Through Plastic Film and Sheeting); ASTM F1249 (Water Vapour Transmission Rate Through Plastic Film and Sheeting); ASTM D2659 (PET and PETG Thermoplastic Containers, where PET layers apply); ASTM D638 (Tensile Properties of Plastics)	DIN EN ISO 527 (Tensile Properties of Plastics); DIN EN ISO 15105 (Gas Transmission Rate Through Plastic Films and Sheets); DIN EN ISO 15106 (Water Vapour Transmission Rate); DIN EN 1186 (Food Contact Materials, where applicable)	FDA 21 CFR 177; EU 10/2011; Barrier Packaging Association Guidelines; UN Packaging Requirements (chemical applications)
Co-extruded Plastic Bottle	IS 14636 (Food Packaging Applications); IS 15609 (Multilayer Packaging Applications); IS 15410; IS 17803; IS 12252; IS 10146; IS 10910; IS 10151	ISO 18601 (Packaging and the Environment); ISO 9001 (Manufacture of Plastic Packaging); ISO 12418-1 (PET bottle post-consumer recycling materials); ISO 13106 (Blow-moulded PP containers and bottle necks)	ASTM D2659 (PET and PETG Thermoplastic Containers, where applicable); ASTM D638 (Tensile Properties of Plastics); ASTM D543 (Resistance of Plastics to Chemical Reagents); ASTM D2463 (Drop Impact Resistance of Blow-Molded Thermoplastic Containers)	DIN EN ISO 527 (Tensile Properties); DIN EN ISO 175 (Chemical Resistance); DIN EN ISO 1183-1 (Density); DIN EN ISO 179-1 (Impact Strength)	FDA 21 CFR 177; EU 10/2011; NSF/ANSI 51 (food applications); UN Packaging Requirements (chemical applications)
Child Resistant Pharmaceutical Bottle	IS 10910 (Polypropylene for its Safe Use in Contact with Foodstuffs, Pharmaceuticals and Drinking Water); IS 7328 ((HDPE) /); IS 12252 (Polyalkylene Terephthalates (PET & PBT) for their Safe Use in Contact with Foodstuffs, Pharmaceuticals and Drinking Water); IS 17803 (Potable Water Bottles (Plastic, Reusable) — Specification);IS 15410; IS 10146; IS 10151	ISO 8317 (Child-resistant Packaging — Requirements and Testing Procedures for Reclosable Packages); ISO 12418-1 (PET bottle post-consumer recycling materials); ISO 13106 (Blow-moulded PP containers and bottle necks)	ASTM D3475 (Classification of Child-Resistant Packages); ASTM D642 (Compressive Resistance of Shipping Containers, where applicable); ASTM D2463 (Drop Impact Resistance of Blow-Molded Thermoplastic Containers); ASTM D638 (Tensile Properties of Plastics)	DIN EN ISO 8317 (Child-Resistant Packaging — Requirements and Testing Procedures); DIN EN ISO 15378 (Primary Packaging Materials for Medicinal Products); DIN EN ISO 527 (Tensile Properties); DIN EN ISO 175 (Chemical Resistance)	US 16 CFR 1700 (Poison Prevention Packaging Act); CPSC Child Resistant Packaging Requirements; FDA 21 CFR 211.132
Tamper Evident Plastic Bottle	IS 14543 (Packaged Drinking Water Bottles) (Tamper-Evident Closure Applications); IS	ISO 21976 (Packaging — Tamper Verification Features for Medicinal Product Packaging); ISO 12418-1 (PET bottle post-consumer recycling materials); ISO 13106	ASTM D3475 (Classification of Child-Resistant Packages, where applicable); ASTM D638 (Tensile Properties of Plastics); ASTM D2463 (Drop Impact Resistance of Blow-Molded	DIN EN ISO 8317 (Child-Resistant Packaging, where applicable); DIN EN 14375 (Child-Resistant Non-Reclosable Packaging for Pharmaceutical Products, where	FDA 21 CFR 211.132 (Tamper-Evident Packaging); USP <671> Packaging Performance; CPSC Packaging Requirements

Product / Application	BIS (IS)	ISO	ASTM	DIN	Other Standards
	14636 (Packaging Applications); IS 15410; IS 17803; IS 12252; IS 10146; IS 10910; IS 10151	(Blow-moulded PP containers and bottle necks)	Thermoplastic Containers); ASTM D4169 (Performance Testing of Shipping Containers and Systems)	applicable); DIN EN ISO 527 (Tensile Properties); DIN EN ISO 1183-1 (Density)	

Source field: Plastic and Plastic Products